

Start

a few offers per device: its thermostat plan, and its plans at the import and export prices

One iteration, repeated

Coordinator LP

meter: import, export, limits, curtailment;
batteries as exact variables
keeps every offer and picks the *blend* with
the cheapest bill plus private costs

prices π , one per slot

offers: plan + private cost

Each device that bids (tank, HVAC, EV)

its best plan if a kWh in slot t costs π_t
it never sees the tariff or the other devices

their answers give a lower
bound; stop when blend –
bound is small

Result

one runnable plan per device (a blend,
where the device can run one)
*certificate: at most the gap from the best
possible*